



SMART INDIA HACKATHON 2026 · INTERNAL ROUND · SOFTWARE

Digital Twin for Well-to-Surface Optimization of **CSS** and **Sucker Rod Pump** Operations

A physics-integrated AI twin for Oil India's Baghewala heavy-oil field, Rajasthan — 8,000–15,000 cP crude, steam cycle and rod pump optimised together in one loop.

PROBLEM ID

SIH26120

ORGANISATION

Oil India Limited

CATEGORY

Software

THEME

Smart Automation

- 17–19° API per PS · 14–17° in field literature

- India's first CSS — BGW-8, Dec 2018

- 218 t → 43,773 t (FY17 → FY26)

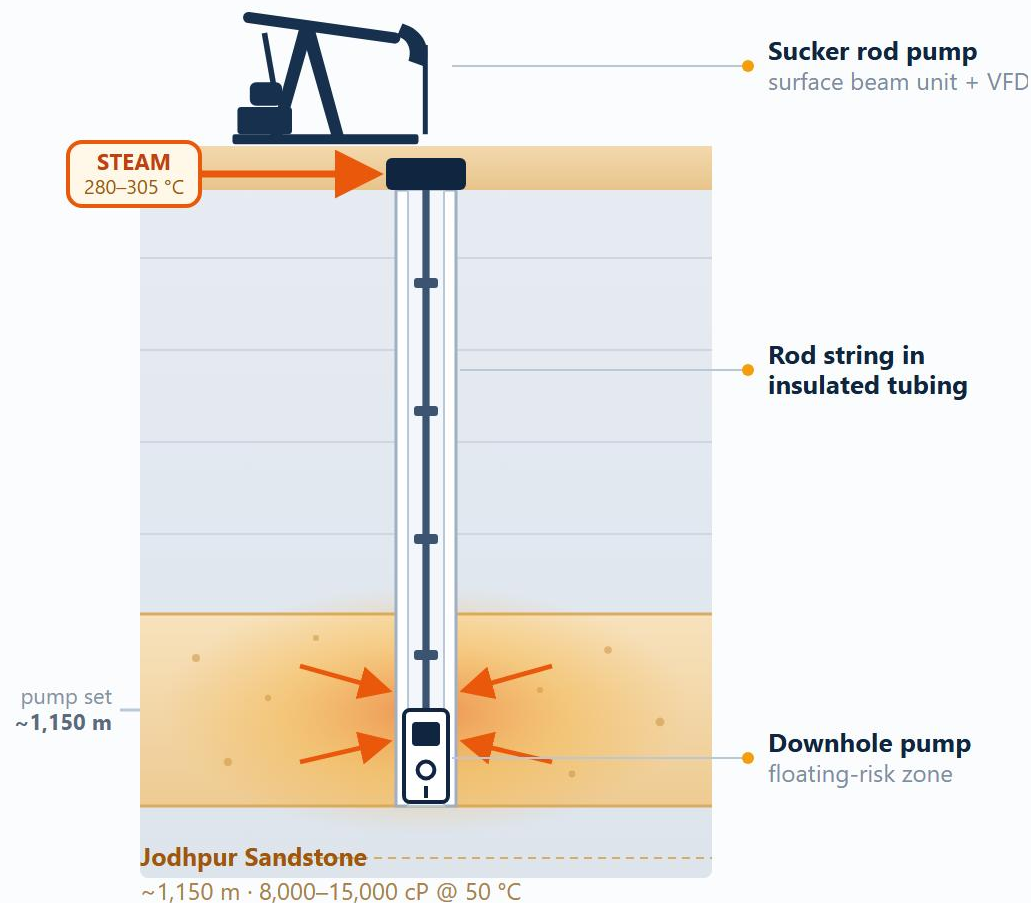
Team **MNIT Jaipur** · Chemical Engineering (lead) + Software · Bikaner–Nagaur basin · ~550–600 km from Jaipur





PROPOSED SOLUTION

Steam and pump, optimised as **one** system — not two



Today: two knobs, two teams

Steam dose and pump speed are set by hand, on separate schedules.



The twin

Heating → viscosity → inflow → rod load, chained in one **sub-second** pass.



The output

Four set-points per cycle: **steam volume**, **soak days**, **cut-off**, **pump speed**.



The guardrail

Nothing is recommended unless rod-floating risk stays in the safe band.

WHY THIS IS NEW

No product co-optimises both halves. Pump optimisers (XSPOC, Lufkin SAM) tune lift alone; thermal-EOR twins (SLB, AVEVA, Kongsberg) tune steam alone.



TECHNICAL APPROACH

Four layers: physics in, **set-points** out

1 DATA**Field & synthetic cycles**

- Oil India Baghewala CSS-cycle logs
- $\geq 3,000$ physics-simulated cycles
- Latin-hypercube sampling, fixed seed

twin/generate_data.py

**2 PHYSICS****Well-to-surface engine**

- Marx–Langenheim steam-zone growth
- Andrade viscosity–temperature law
- Vogel IPR + rod load & pump efficiency

twin/thermal.py · srp.py

**3 LEARNING****Surrogate + optimiser**

- XGBoost SOR surrogate on physics runs
- Rod-floating risk classifier
- Bayesian search under safety constraints

ml/train.py · ml/optimize.py

**4 DELIVERY****Twin dashboard & API**

- FastAPI /simulate /optimize /params
- SOR gauge, dynamometer card, alerts
- One recommendation card per cycle

api/main.py · dashboard/

🔄 Closed loop — accepted set-points feed back as new training rows: VFD speed & steam-control schedule → next cycle

THERMAL**Marx–Langenheim (1959)**

$$A_s(t) = (Q_i M h) / (4 k_{ob} \rho c_{ob} \Delta T) \cdot F(t_D)$$

How far the heated zone spreads per unit steam.

FLUID**Andrade viscosity law**

$$\mu(T) = A \cdot \exp(B / T)$$

Fitted to Baghewala's own μ -T data.**INFLOW****Vogel IPR (1968)**

$$q/q_{\max} = 1 - 0.2(p_{wf}/\bar{p}) - 0.8(p_{wf}/\bar{p})^2$$

What the heated reservoir delivers to the pump.

STACK

Python 3.12

NumPy · SciPy

XGBoost

scikit-optimize

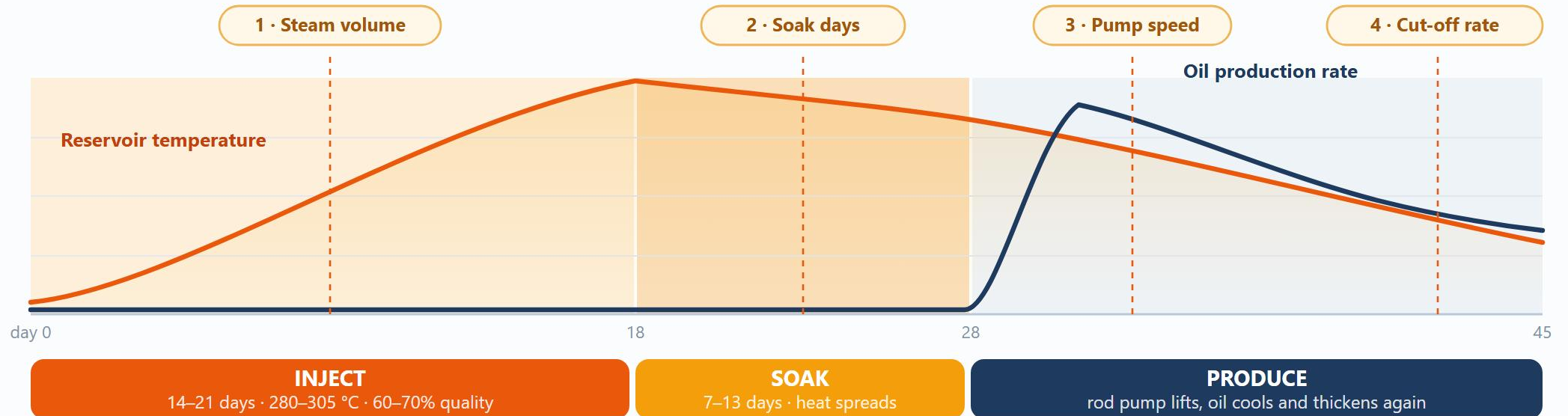
FastAPI

Plotly dashboard

pytest — physics unit tests



What the twin actually decides, **cycle by cycle**



FOUR CONTROL VARIABLES — BOUNDED BY REAL FIELD PRACTICE

- 1 Steam volume** Generator-limited over a 14-21 day window
- 2 Soak duration** Field practice: 50-60% of the injection phase
- 3 Pump speed (SPM)** VFD range, capped by the rod-floating limit
- 4 Cut-off rate** When the cycle stops paying — steam goes back in

OBJECTIVE & CONSTRAINT

minimise $SOR(x)$ $t_{steam} / m^3 \text{ oil}$
s.t. $P(\text{rod float} | x) < 0.30$
 $x = [V_{steam}, t_{soak}, N_{pump}, q_{cut}]$

gp_minimize with 10 random restarts; floating index > 0.6 is treated as unsafe and rejected.

Benchmark: heavy-oil CSS runs at an SOR of roughly 3-8 (average ~6), efficient below 3. Bagheewala's own SOR is not publicly reported.



FEASIBILITY & VIABILITY

Settled physics, real parameters, a **de-risked** field



The physics is settled

Marx–Langenheim (1959), Andrade / ASTM D341, Vogel (1968) — textbook models, 50+ years of use. No research risk, only integration.



The parameters are real

`field_params.json` is calibrated to SPE and Oil India data: depth ~1,150 m, μ 11,500 cP @ 50 °C, steam 290 °C, porosity 0.09.



The field is de-risked

India's first CSS ran here — BGW-8, Dec 2018 — with a 5–6× first-cycle uplift. 52 wells drilled, 33 operational.

BUILD PATH — UNDER FOUR WEEKS TO A LIVE API

WEEK 1

1

Physics engine

thermal · viscosity
IPR · rod dynamics

WEEK 2

2

Synthetic cycles

≥ 3,000 runs, seeded
physics unit tests

WEEK 3

3

ML + optimiser

XGBoost surrogate
Bayesian search

WEEK 4

4

API + dashboard

FastAPI endpoints
live Plotly twin

FINALE



Oil India data

re-fit constants,
validate on history

RISK 01 · DATA

Oil India history arrives late

Physics-generated cycles train the model today; field data only re-fits constants, it is not a blocker.

RISK 02 · UNCERTAINTY

Reservoir properties vary

One source of truth in `field_params.json`, with Monte-Carlo sensitivity sweeps in the test suite.

RISK 03 · OPTIMISER

Search finds a local minimum

Ten random restarts, plus hard physics bounds — injection pressure stays under the fracture gradient.

Porosity below 10%, poor-to-fair reservoir quality — a genuinely hard target. That difficulty is the argument for a twin: trial-and-error costs a full 30-day cycle per experiment.



IMPACT & BENEFITS

Less steam per barrel, fewer **rod-pump failures**

20–30%**Targeted SOR reduction**

Physics-simulated and internally consistency-checked. Field validation is the next phase.

\$15k–50k**Saved per avoided workover**

Industry-typical cost of one sucker-rod-pump failure: rig time, rods, lost production.

52 / 33**Wells drilled / operational**

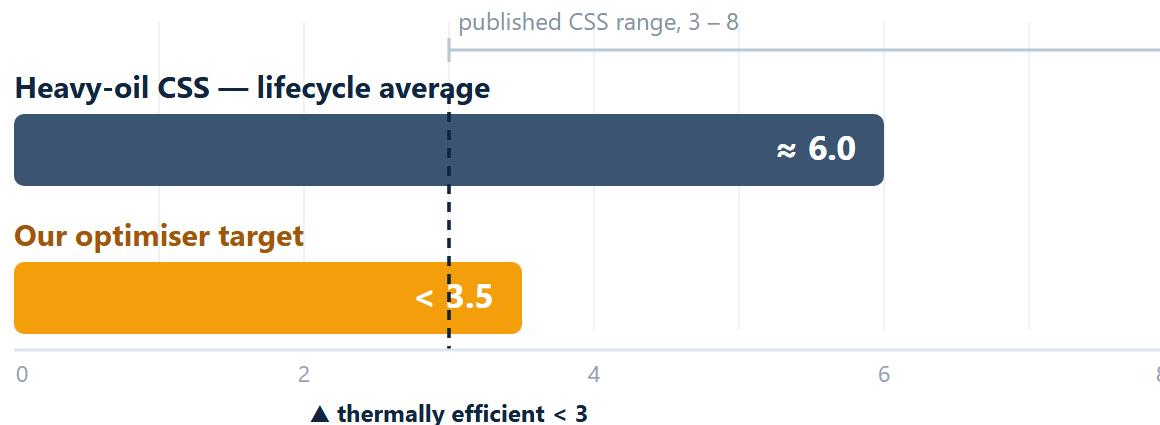
19 CSS'd in FY2025-26 alone — ~72% more than the year before.

~200×**Output growth since CSS**

218 t (FY17) → 43,773 t (FY26).

**STEAM-OIL RATIO — WHERE WE ARE AIMING**

tonnes of steam per m³ of oil produced — lower is better



Illustrative — to be validated on Oil India data. Baghewala's own SOR is unreported.

**Economics**

Less steam per barrel, and every avoided rod-pump workover is **\$15k–\$50k** not spent.

**Safety**

Flags metal-on-metal rod and tubing wear — **over half of all SRP failures** — before it happens.

**Sustainability**

Steam here costs ~3.0 GJ of diesel per tonne. Less steam is directly less CO₂ per barrel.

**Scalability**

Swap `field_params.json` and the same twin moves to the next CSS well, or the next field.

Stated honestly: the 20–30% figure is what our physics model predicts, not a measured field result — and no field failure-reduction percentage is claimed, because that needs a pilot. Exact ₹/\$ per well needs Oil India's own cost data.



RESEARCH & REFERENCES

Every number traces to a source we can show you

PHYSICS FOUNDATION — PEER-REVIEWED

- ML** **Marx, J. W. & Langenheim, R. H.** (1959). Reservoir heating by hot fluid injection. *Trans. AIME*, 216, 312–315.
Used for: steam-zone growth and conduction loss to the overburden
- A** **Andrade, E. N. da C.** (1934). The viscosity of liquids. *Nature*, 125, 309–310 — with **ASTM D341** for viscosity–temperature charts.
Used for: $\mu(T) = A \cdot \exp(B/T)$, fitted to Baghewala's own μ – T anchor
- V** **Vogel, J. V.** (1968). Inflow performance relationships for solution-gas drive wells. *J. Petrol. Tech.*, 20, 83–92.
Used for: the IPR curve — flow rate against pressure drawdown

FIELD & INDUSTRY EVIDENCE

- SPE** **SPE-23APOG-535203** (2023). Enhancement of production of heavy and highly viscous crude using an electrical downhole heater.
Used for: Baghewala depth, API gravity, viscosity
- OIL** **Oil India Limited** — Rajasthan field pages and operations data for well **BGW-8**.
Used for: steam rate, 280–305 °C, 85–97 kgf/cm², 60–70% quality, well counts, output history
- OGJ** **Oil & Gas Journal** — "OIL starts CSS of well in Rajasthan."
Used for: India's first CSS, Dec 2018, Belgrave Oil & Gas partnership
- SCAN** **Competitive landscape scan** — XSPOC, Lufkin SAM/SROD, Weatherford ForeSite, SLB OptiSite, AVEVA, Kongsberg Kognitwin.
Used for: confirming that none of them co-optimize CSS with rod lift

MNIT

MNIT Jaipur — thermodynamics & fluid mechanics labs

Departmental precedent for validating the physics models, and the ChemE supervision behind this build.

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unverifiable citations